

of Azure's VM cores and 60 per cent of the images on it are Linux based. In fact, more than half of the ecosystem at Microsoft is based on Linux. SaaS and various other applications are leveraging open source too.

He says, "Any new offering coming to the market can continue to leverage its core DNA while also getting exposed to the larger market worldwide. This gives an opportunity to experience the full might of larger cloud providers."

Solving a problem using open source can help solve many other problems. For example, a normal customer service app has to touch different platforms, and be compatible with different types of devices. It may also have to browse through multiple databases. The idea, says Sharma, is to slowly and steadily multiply what is being offered to clients.

"You begin by offering one solution to one type of customer base and then you can pivot to a larger market. Today's cloud provides you that opportunity to experiment," he adds.

Microsoft had acquired GitHub for US\$ 7.5 billion back in 2018. Then, in March 2021, it announced a grant of Rs 10 million to fund India's developers who are building software using open source tools.

Subsidise and maximise

It is pretty clear that the open source business model of tech giants like Microsoft, Amazon and Google is subsidised by the cloud services they sell. Facebook, as per Hariri, is subsidised by data analytics and advertising. Organisations like JetBrains subsidise the business of open source by selling tools.

"I don't think there are any key parameters. You should look at how you are going to sustain your open source contributions. The best example that can be thought of here is Red Hat," explains Hariri.

Picking the right open source business model

"You can always look at it as a half-full or half-empty glass. We can leverage a larger ecosystem. We can reach out to companies all over the world by leveraging the public clouds."

— **Sanjeev Sharma**, director, new ISV partnerships, Microsoft

"Focus more on the pain of the consumer, and then make a business model that can help you survive. Times are changing rapidly. It's good to be on the dynamic side, as these changes will help you survive in the long run."

— **Samir Doshi**, co-founder and director at iNextrix Technologies

"It's important to keep figuring out things on the way. It's critical to always remain agile and flexible. Everyone should nurture their curiosity and keep testing waters to see what works best for them. The market is optimistic; so why shouldn't you be."

— **Kabir Chandhoke**, COO, SourceFuse Technologies

"If you want to leverage open source, you should first mull over your decision. Think about whether open source will provide any benefits or not. Once you are certain, make sure you develop a business model around it. If you want to make it successful, make sure you have a licence that is not overly permissive."

— **Hadi Hariri**, VP, developer advocacy, JetBrains

The Red Hat business model, as most industry *gurus* will agree, is complicated yet successful enough to require consulting, which is provided by the company team as part of the model. This business model may become less sustainable if the consulting is taken care of by a third party.

Open source, as Mahajan points out, might be more of an enabler that helps get traction and build newer business models. What helps new business models based on open source to sustain is the demand they get from different business verticals.

"When running a business, sometimes you need to subsidise what you are doing for the greater good," says Kabir Chandhoke, COO, SourceFuse Technologies.

Marketplace, enterprise and community software; and the role of the cloud

Most open source business models in use today are offered as enterprise and community editions. The community edition provides free usage of open

source with limited capabilities, whereas the enterprise edition offers a lot more features at a cost. "These two models are becoming very popular. On top of these, you can also offer complimentary services such as certification, development services and support contracts," says Samir Doshi, co-founder and director at iNextrix Technologies.

But it's open source marketplace software that joins all the dots together.

"Marketplace is definitely a brilliant business model, as it adds a lot of value to all the open source products. You have brilliant brains coming up with add-ons and applications," says Doshi.

It is critical to understand the relationship between open source and the cloud in order to understand business models better. The debate around whether the cloud is making open source popular, or vice-versa, is something that experts still lock horns on.

"Open source is what actually drove the cloud in the first place. Can you imagine the public clouds we have today and companies paying licence fees for their operating systems?" asks Chandhoke. "We would never have